

Utsab Karan

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EDUCATION

Indian Institute of Technology Kharagpur

Integrated Masters in Electronics and Electrical Communication Engineering; GPA: 9.50/10

Kharagpur, India

Aug. 2023 – Present

EXPERIENCE

Autonomous Ground Vehicle Research Group

Vision Head | Guide: Prof. Debashish Chakravarty

IIT Kharagpur

Apr. 2024 – Present

- Deployed **monocular depth** estimation models and compared outputs with Intel RealSense **stereo depth**.
- Pipelined ArUco marker detection, grid mapping and autonomous navigation for **TurtleBot** simulations.
- Developed **Pure Pursuit** and **Stanley** controllers for F1tenth racing simulations.
- Tested RL algorithms including **DQN**, **A2C** and **PPO** on OpenAI Gymnasium environments.

Autonomous Robots and Multi-robot Systems Lab, SysCon

Research Intern | Guide: Prof. Leena Vachhani

IIT Bombay

May 2025 – Dec. 2025

- Simulated **social crowd** environments using Gazebo and ROS2.
- Analysed various scenarios and **compiled datasets** for human trajectory modelling interaction
- Worked on gap-based **topological mapping** with safe kinodynamically feasible planning and control.
- Implemented and tested algorithms on the **TurtleBot3** burger.

Vision and Artificial Intelligence Lab

Research Intern | Guide: Prof. Venkatesh Babu

Indian Institute of Science

May 2026 – Present

- Benchmarked data pruning strategies on **ImageNet** and **CIFAR-100** using ResNet architectures.
- Working on **Continual** and **Data Efficient Learning** methods for computer vision.

PUBLICATIONS

- K. Trivedi, U. Karan, A. Sudarshan and L. Vachhani, "Safe Indoor Exploration and Sparse Topological Mapping with Integrated Planning and Control," *2025 Eleventh Indian Control Conference (ICC)*, Bangalore, India, 2025. [[Paper](#)]
- Moshayedi AJ, Roy AS, Karan U, Zhang M, Bassir D. *An Integrated Beetle Antennae Search-Enabled Navigation Framework for Omnidirectional AGV Mobile Robots in Unknown Environments*. **EAI Endorsed Trans AI Robotics** [Internet]. 2026 Feb. [[Paper](#)]

PROJECTS / COMPETITIONS

AlphaZero Implementation for the Black-Hole Game [[Link](#)]

The KGPian Game Theory Society

IIT Kharagpur

Jan. 2026 – Feb. 2026

- Deployed classical RL **self-play** to train the agent in Pytorch.
- Implemented the AlphaZero style framework with Monte-Carlo Tree Search guided decision making.

Diffusion Language Model from Scratch [[Link](#)]

Self Project

IIT Kharagpur

Dec. 2025 – Present

- Implemented a diffusion-based **BERT** style 15M parameter language model in PyTorch.
- Worked on posterior regularization, x_0 -parameterized denoiser and clamping-based decoding.
- Training and experimenting on **ROCStories** and **E2E** datasets.

Monocular Depth Estimation for Edge Devices [[Link](#)]

Autonomous Ground Vehicle Research Group

IIT Kharagpur

Sept. 2025 – Dec. 2025

- Implemented a lightweight **encoder-decoder** depth and segmentation model in PyTorch.
- Used pixel shuffle, inverted residual bottlenecks and skip connections for efficient decoding.
- Trained and benchmarked on **KiTTi** and **Cityscapes** datasets.

Beetle Antenna Search Optimisation for Robotic Navigation [[Link](#)]

Remote Collaboration | Prof. Ata Jahangir Moshayedi

Oulu University

June. 2025 - Aug. 2025

- Implemented the Beetle Antenna Search(BAS) algorithm for optimising **PID** based controls.
- Tested BAS optimised **A*-path planning** for obstacle avoidance.
- Tested and compared robustness of solution across different environments.

Pure Pursuit Controller — F1Tenth [\[Link\]](#)

Autonomous Ground Vehicle Research Group

IIT Kharagpur

Aug. 2024 – Sept. 2024

- Designed a Pure Pursuit controller for **F1Tenth** racing simulations.
- Evaluated performance across multiple tracks using **RViz** and large waypoint datasets.
- Integrated ROS libraries, transformations, message formats and node parameters.

COURSEWORK

Department Courses: Advanced Calculus, Numerical Analysis, Probability and Statistics, Linear Algebra, Optimization Methods, Analog Circuits, Digital Electronics, Network Theory, Signals and Systems, Systems and Control, Electromagnetic Engineering, Communication-I, Communication-II, Digital Signal Processing - I, Digital Signal Processing - II, Algorithms, RF and Microwaves, VLSI Design, Computer Architecture

Electives: High Performance Scientific Computing, Information Theory and Coding

Others: Game Theory-I (Stanford), Programming for Robotics (ETH Zurich), Supervised ML

COURSE PROJECTS SUMMARY

Communications Lab: Designed circuits for realizing various signal modulation techniques. **Digital Signal Processing Lab:** Implemented FIR and IIR Filters. Designed a Dual Tone Multifrequency (DTMF) Coder & Decoder in Matlab. **DIY Lab:** Built an Object Tracking Aim Bot using OpenCV and IoT integration. **High Performance Scientific Computing:** Parallelized Sparse Matrix Solvers using OpenMP, MPI and CUDA. **Embedded Systems Lab:** Designed a real-time clock using Assembly language and implemented on the 8051 test board. **VLSI Design Lab:** Designed and Verified Layouts for Gates and Ring Oscillators.

TECHNICAL SKILLS

Languages: Python, C/C++, CUDA C, Arduino, Verilog, OpenMP, MPI

Libraries: NumPy, Pandas, PyTorch, Matplotlib, OpenAI Gymnasium, D4RL

Frameworks/Software: ROS1/ROS2, Gazebo, RViz, Vivado, LTSpice, Simulink, Docker, Git, Linux

AWARDS AND ACHIEVEMENTS

- Rank **15** among 124k candidates in **WBJEE 2023**
- Rank **1784** with 99.86 percentile score in **JEE Mains 2023**
- Rank **1596** with 99.16 percentile score in **JEE Advanced 2023**
- **4th** place in Hardware Modeling in **General Championship 2025** [\[Link\]](#)
- **1st** in Qualification and **9th** Globally — Roboracer Sim Racing League @ **ICRA 2025** [\[Link\]](#)
- **2nd** in Robust Visual Foraging Competition @ **NeurIPS 2025** [\[Link\]](#)
- **5th** in Qualification and **7th** Globally — Roboracer Sim Racing League @ **CDC-TF 2025** [\[Link\]](#)